HASSAN SHAHID

Agentic AI Engineer / Data Scientist

https://hasanshahidd.github.io/portfolio/ Lahore, Pakistan | +92-336-7798256 | ch.hassanshahid3@gmail.com

SKILLS

- **GenAI & LLM Engineering:** Hands-on experience building GenAI pipelines with LangChain, AutoGen, CrewAI, and ReAct-style reasoning; implemented tool use, memory, and RAG over vector stores (Pinecone, FAISS, Chroma), STT/TTS
- **Enterprise AI Solutions:** Built enterprise assistants for GRC, banking compliance, procurement, and document intelligence with guardrails and validation; expert in prompt design, chaining, multi-agent orchestration, and fallback strategies
- **LLM Integration:** Integrated GPT-4/4o, Claude-family models, and open-source LLMs for compliance Q&A, reporting, and decision support; designed agent-based pipelines that translate natural language into safe, structured actions
- **Python & ML Stack:** Proficient in Python (FastAPI, Flask, Dash) with deep use of pandas, NumPy, scikit-learn, TensorFlow, PyTorch, and NLP tooling; built production APIs, async services, and inference endpoints
- **Data Engineering:** Designed production-grade ETL pipelines using Python and Airflow; automated ingestion from APIs, databases, and documents; implemented OCR-driven workflows (Tesseract) with downstream LLM reasoning
- **Cloud & DevOps:** AWS, GCP, Azure; Docker, Kubernetes, CI/CD pipelines; experience deploying and scaling production AI services across cloud environments
- **Analytics & Visualization:** Built decision-critical dashboards in Power BI, Dash, and Plotly; strong SQL skills across relational databases; experience with Snowflake and enterprise data systems
- **AI Safety & Reliability:** Confidence scoring, hallucination mitigation, role-based access control, human-in-the-loop escalation, audit-ready logging, and tenant isolation across enterprise deployments
- **Work Model:** Flexible work model; open to async remote engagements across EU/UK/US/AUS time zones; comfortable with trial-based roles and agile delivery

WORK EXPERIENCE

AI/ML Engineer (GenAI & Enterprise Systems)

Liztek | *September 2025 - Present*

- Architected and delivered AI-powered GRC and procurement assistants, enabling auditors, risk managers, and procurement teams to query policies, controls, vendors, and risks via natural language interfaces.
- Led a banking-sector engagement with State Bank of North America - built a compliance intelligence layer over their regulatory schema, automating control mapping, audit evidence retrieval, and regulatory reporting with LLM-powered RAG pipelines.
- Designed safe multi-table agentic query pipelines across complex enterprise schemas with full role-based access control and tenant isolation, cutting audit preparation time by 60-70%.
- Integrated LLM agents with backend APIs for risk summaries, compliance dashboards, and policy gap analysis; implemented guardrails and confidence scoring to meet enterprise and regulatory standards.
- Collaborated cross-functionally with product, compliance, and engineering teams to align AI outputs with real enterprise workflows and audit requirements.

Data Scientist and ML Engineer (Leading GenAI)

Optivex Solutions (greeka) | Lahore, Pakistan | *April 2024 - July 2025*

- Led a team of 5 analysts to build enterprise-wide KPI dashboards used by senior leadership for weekly decision-making across sales, product, and marketing.
- Analyzed 10M+ data rows across marketing, finance, and ops; delivered strategic insights that contributed to a 15% revenue lift and quarterly C-level planning.
- Initiated and delivered an LLM-powered business intelligence reporting engine using LangChain, agents summarise KPI data, surface anomalies, and auto-generate executive briefings.

- Designed and deployed an OCR-powered text extraction and auto-summarization pipeline using Tesseract and LangChain, improving document processing speed by 50%.
- Automated reporting pipelines with Python and Excel macros; reduced manual reporting time by 40% and delivered executive-ready reports in near real-time.

Sales Analyst/Scientist

Fierra Footwear | Lahore, Pakistan | *January 2023 - February 2024*

- Led a 5-member analytics team on sprint-based retail analysis; designed Power BI dashboards used by C-level to optimize inventory strategy, reducing overstock by 25%.
- Used SQL and Excel to extract insights from category-wise sales data across 10+ product lines and increased monthly revenue by 12%.
- Applied GenAI models to generate personalized mobile ad creatives based on past customer behavior, improving engagement metrics by 22%.

KEY PROJECTS

Agentic Procurement Intelligence Platform

CrewAI · LangChain · GPT-4o · FastAPI · Pinecone · AWS

- Architected a fully autonomous multi-agent procurement workflow using a 4-agent CrewAI crew (Intake, Vendor Intelligence, Approval Orchestrator, Audit Trail), enabling natural language PO resolution with zero manual triage.
- Built Pinecone vector store over 50K+ procurement docs for real-time clause retrieval and contract risk flagging; integrated live ERP APIs with guardrails for spend-threshold enforcement and human-in-the-loop escalation for high-value decisions.
- Reduced procurement cycle time by 55% in pilot; deployed on AWS Lambda + API Gateway with FastAPI backend and audit-ready logging, fully compatible with enterprise GRC and SOX compliance requirements.

Enterprise GRC & Compliance Intelligence (Senior-Level Contribution)

LangChain · GPT-4o · RAG · Pinecone · FastAPI

- Led end-to-end development of an AI-driven GRC platform enabling auditors and compliance teams to query controls, risks, and policies via natural language across complex enterprise schemas using LLM-powered RAG pipelines with validation layers and confidence scoring.
- Banking-sector deployment with State Bank of North America automated regulatory control mapping, compliance gap analysis, and audit evidence generation - cutting manual audit prep time by 60–70%.
- Built role-based access controls, tenant isolation, and explainability traces across multi-client deployments; improved audit decision turnaround by 40%.

Voice AI Patient Registration System

Vapi.ai · GPT-4o-mini · FastAPI · PostgreSQL · Railway

- Built a production-ready voice AI agent answering real U.S. phone calls, conducting natural LLM-driven conversations to collect patient demographics, validate input in real time, and persist confirmed records to PostgreSQL with duplicate detection and soft-delete audit trail.
- Architected a 4-layer pipeline: Vapi.ai (telephony + STT/TTS) - GPT-4o-mini (conversation logic) - FastAPI (validation + business logic) - PostgreSQL; deployed on Railway with live REST API, full CRUD, and a web dashboard for record management.
- Implemented UUID primary keys for horizontal scalability, async endpoints for concurrent call handling, and HIPAA-aware data design with no audio persistence - Deployed

OMANI Therapist Voice

Azure STT/TTS · GPT-4o · Claude Opus 4 · Groq · Flask · Docker · Hugging Face

- Engineered a real-time, voice-only mental health chatbot delivering culturally sensitive therapy in Omani Arabic, achieving sub-20s end-to-end latency (avg. 12–15s) via a pipeline of Azure STT - dual-model emotion detection - safety screening - GPT-4o/Claude Opus 4 response - Azure TTS.
- Implemented a fine-tuned Hugging Face emotion model on annotated Omani Arabic phrases with CBT-based therapy response generation, crisis keyword safety screening with escalation protocols, and cultural embeddings for authentic dialect adaptation.
- Deployed on Docker at zero monthly cost using Azure Free Tier (STT/TTS) and Groq free-tier LLM inference; designed with HIPAA-aware logging, no audio persistence beyond immediate processing, and Kubernetes-compatible architecture for production scaling.

Real-Time Healthcare Readmission Risk Prediction (ML + GenAI)

XGBoost · Dash · Flask · Plotly · scikit-learn

- Engineered a predictive XGBoost model on a 1M-row dataset with 5-fold cross-validation; automated risk scoring for 200K test patients, reducing readmissions by 15% and saving $3.5M annually.
- Built a real-time Dash dashboard with Flask API enabling 30-second updates across 6 visualizations for 1K daily patient queries, cutting intervention delays by 30% and saving $1.2M in operational costs.
- Identified a 15% winter readmission spike in the 61-80 age group across 600K records, optimizing resource allocation to reduce peak-season readmissions by 10% and saving an additional $2M.

EDUCATION

Bachelor of Science in Artificial Intelligence

Superior University, Lahore

- Relevant Courses: Data Science, Programming with AI, Python Programming, Artificial Intelligence
- Certifications: Big Data Analytics, Python Wrangling, Python with AI